

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

COMPUTER SCIENCE

Examination, May 2020

Nature-Inspired Computation

Module Leader: Dr. Alberto Moraglio

Duration: TWO HOURS + 30 MINUTES UPLOAD TIME

Answer ALL questions.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

This is a OPEN BOOK examination.

Question 1

- (a) What are the main algorithmic elements of *particle swarm optimisation* (PSO)? What are the main parameters of PSO and feasible parameter settings?

(7 marks)

- (b) Briefly describe the main elements of an algorithm to simulate the collective behaviour of birds. Provide the canonical name of such an algorithm.

(7 marks)

- (c) There are many different variants of neural networks that exist in addition to the most popular type, the multilayer perceptron. For each of the following networks, briefly explain the key differences between the network provided and a multilayer perceptron in terms of their architecture and the type of data they would be best suited to process.

- (i) Perceptron.

(2 marks)

- (ii) Self organising map.

(2 marks)

- (iii) Recurrent neural network.

(2 marks)

- (d) Optimisation algorithms must traverse a *fitness landscape*, that will be different for each problem encountered.

- (i) Explain what is meant by the fitness ‘landscape’.

(3 marks)

- (ii) Explain why a simple algorithm such as a hillclimber is unlikely to find a good solution in complex search landscapes.

(3 marks)

- (e) Explain the notion of *emergence* in the context of Conway’s “Game of Life”.

(6 marks)

- (f) Provide an example of a geometric object whose dimension is intermediate between *two* and *three*. Explain why this is so.

(3 marks)

ECM3412

- (g) What are the steps to build the Koch curve? What is the length of the Koch curve?

(5 marks)

(Total 40 marks)

Question 2

Ant colony optimisation can be used for a number of different applications.

- (a) Describe the main algorithm and, briefly, the experimental findings in the paper Di Caro G., Dorigo M. (1998) “Ant Colonies for Adaptive Routing in Packet-Switched Communications Networks” in the Proceedings of PPSN – Lecture Notes in Computer Science Vol. 1498, pp. 673-682.

(15 marks)

- (b) Ants in ant colony optimisation (ACO) algorithms need a topology to traverse to solve an optimisation problem.

- (i) Explain what is meant by a construction graph in ant colony optimisation.

(5 marks)

- (ii) Draw a construction graph for a problem with 5 variables and 3 choices (decisions) per variable.

(5 marks)

- (c) Describe the experiments you would conduct to test the ACO formulation above. In your answer pay particular attention to the parameter settings that you would test and the experimental conditions used.

(5 marks)

(Total 30 marks)

Question 3

Evolutionary algorithms are a family of general problem solving methods that use the principle of evolution as their basis.

- (a) What are the main common algorithmic elements of *evolutionary algorithms*? What are their main parameters and feasible parameter settings?

(6 marks)

- (b) What are the main similarities and differences between *genetic algorithms* and *genetic programming*?

(6 marks)

- (c) What is the difference between *direct* representation and *indirect* representation in evolutionary algorithms? Give an example of each in the context of timetabling contrasting the two.

(6 marks)

- (d) Multi-objective evolutionary algorithms differ in several ways from their single objective counterparts.

- (i) Explain what is meant by the term Pareto-front in multi-objective evolutionary algorithms. Give an example of a Pareto-front.

(6 marks)

- (ii) Explain what the desirable properties of the Pareto-front are and why these properties lead to better decision making for the end-user

(6 marks)

(Total 30 marks)